

1. Experimental acquisition

Box–Behnken design
(T , solvent, solid–liquid ratio)

UV–Vis determination
(triplicate absorbance)

Reference concentration–time profiles

2. Grey-box kinetic modeling

Consecutive Ext–Deg model fit
(estimate k_{ext} , k_{deg})

Physical validity filter
($R^2 \geq 0.50$)

3. Physics-informed augmentation

Monte Carlo augmentation
(noise level σ)

Phenomenological variance filter
($\pm 60\%$ deviation)

k_{ext} with 95% confidence intervals

4. Predictive mapping & validation

Random forest mapping
(operating variables $\rightarrow k_{\text{ext}}$)

Out-of-condition (grouped) validation
vs. in-condition & baselines

Gaussian process uncertainty map